Using Dynamic Non-Dominated Sorting Genetic Algorithm to Optimize Vehicle Locomotion Along Varying Curves

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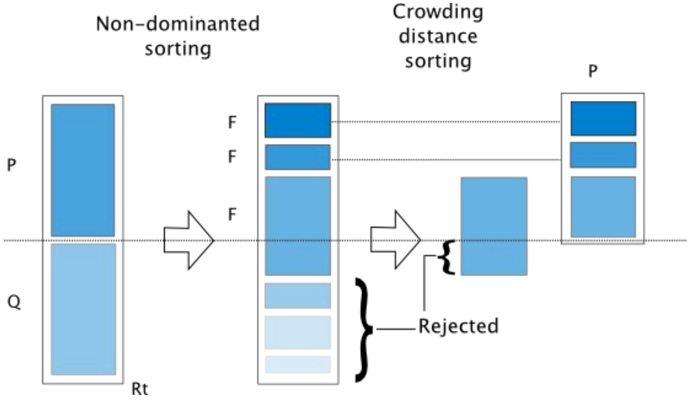
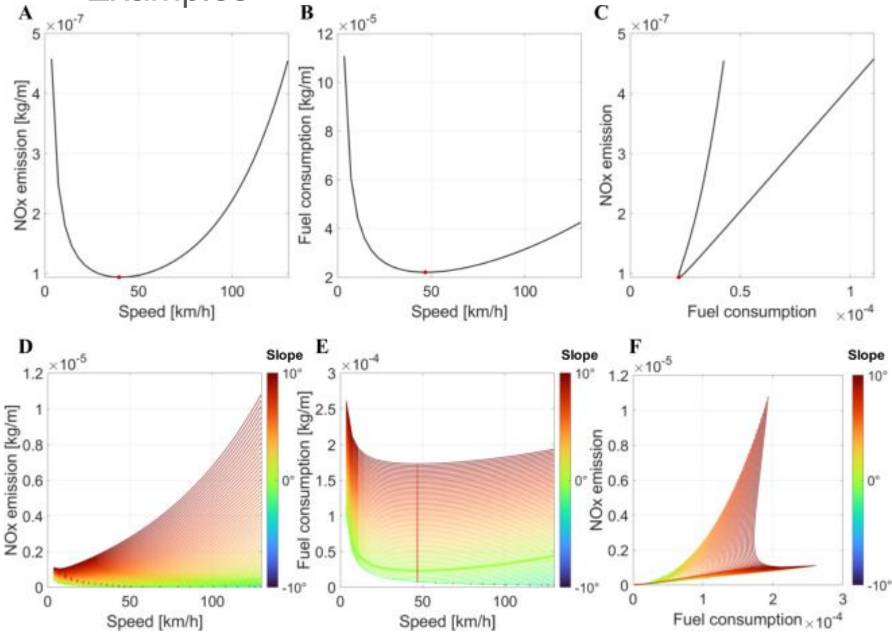


Background

- Autonomous vehicles need adaptability
- Multiple objectives must balance
- Speed vs. stability tradeoff
- Robust systems improve safety

Examples



Research Question

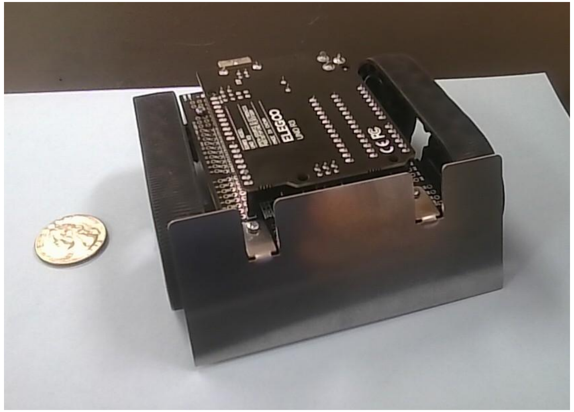
- Can NSGA-II improve driving?
- Can dynamic optimization increase robustness?
- How will a vehicle using this algorithm respond to varying track conditions?

Hypothesis

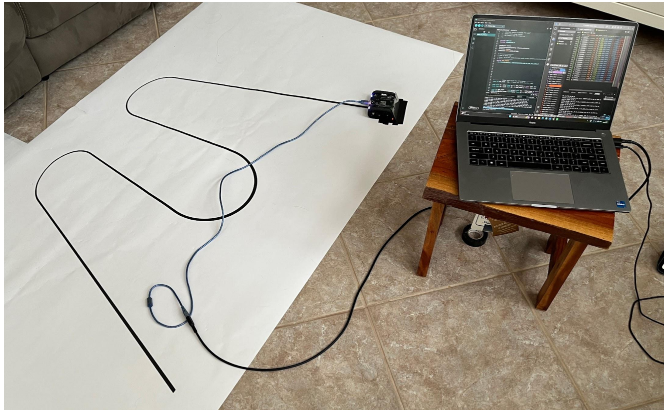
- Dynamic NSGA improves adaptability
- Pareto optimization balances objectives
- Leads to faster times with lower veer

Materials

- Micromouse robot
- Arduino UNO Rev3
- Plotter printer
- Pegboard testing surface
- Laptop and batteries



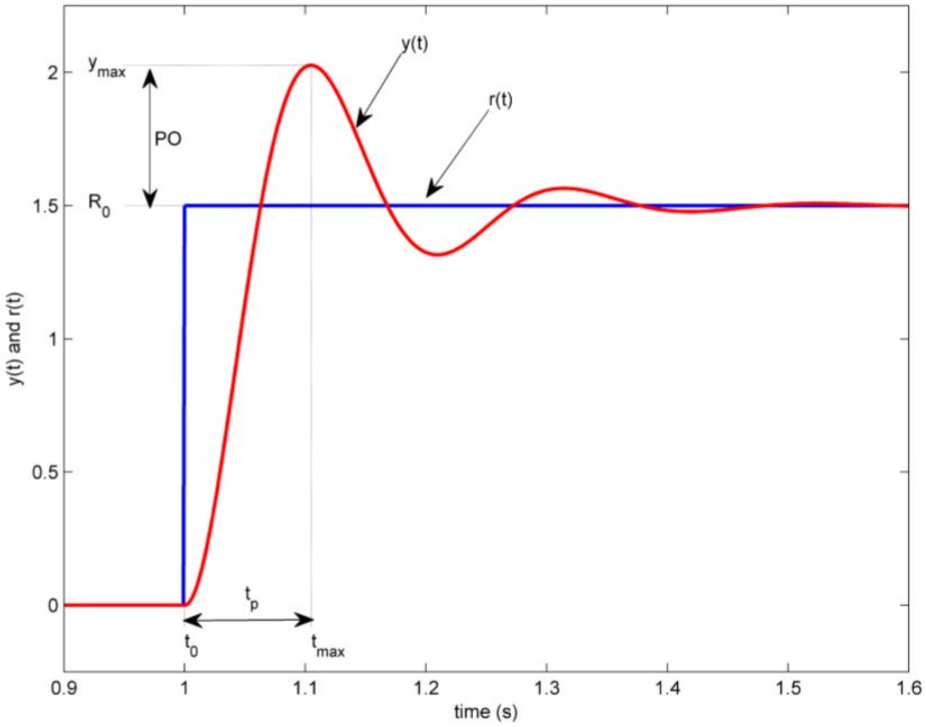
Micromouse (Quarter for reference)

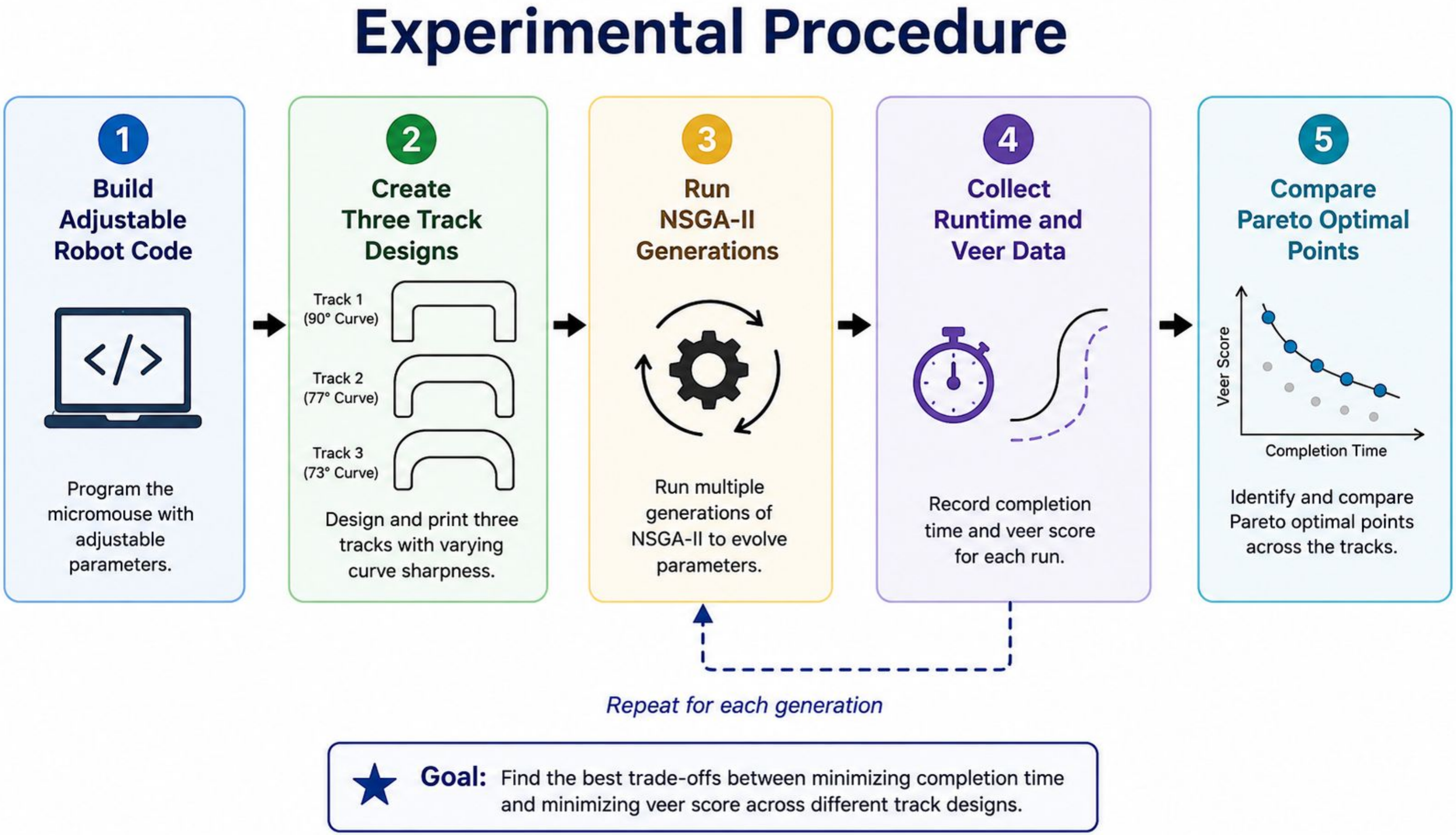


Testing Setup

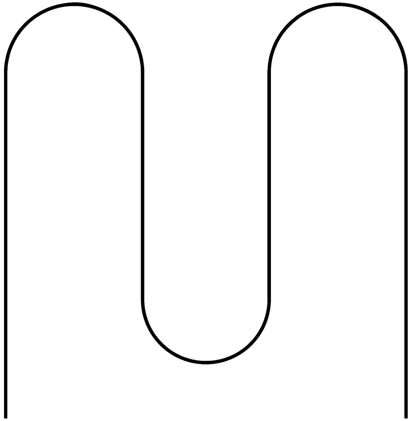
Variables

- kp proportional gain
- kd derivative gain
- Base speed
- Minimum speed
- Brake power
- Curve stutter settings



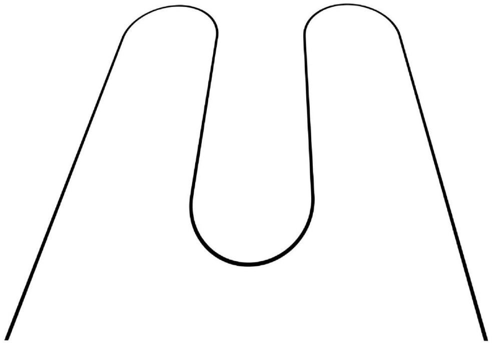

Experimental Procedure
1
Build Adjustable Robot Code
Program the micromouse with adjustable parameters.
2
Create Three Track Designs
Track 1 (90° Curve)
Track 2 (77° Curve)
Track 3 (73° Curve)
Design and print three tracks with varying curve sharpness.
3
Run NSGA-II Generations
Run multiple generations of NSGA-II to evolve parameters.
4
Collect Runtime and Veer Data
Record completion time and veer score for each run.
5
Compare Pareto Optimal Points
Veer Score
Completion Time
Identify and compare Pareto optimal points across the tracks.
Repeat for each generation
Goal: Find the best trade-offs between minimizing completion time and minimizing veer score across different track designs.

Track 1



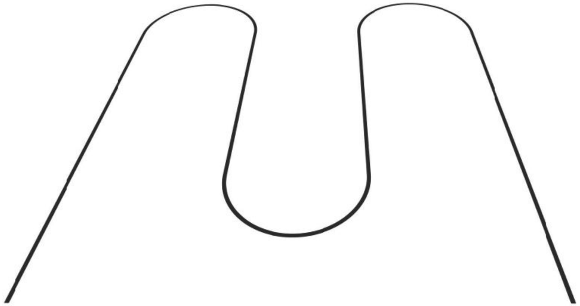
Regular entry curve of 90°

Track 2



Entry curve of about 77°

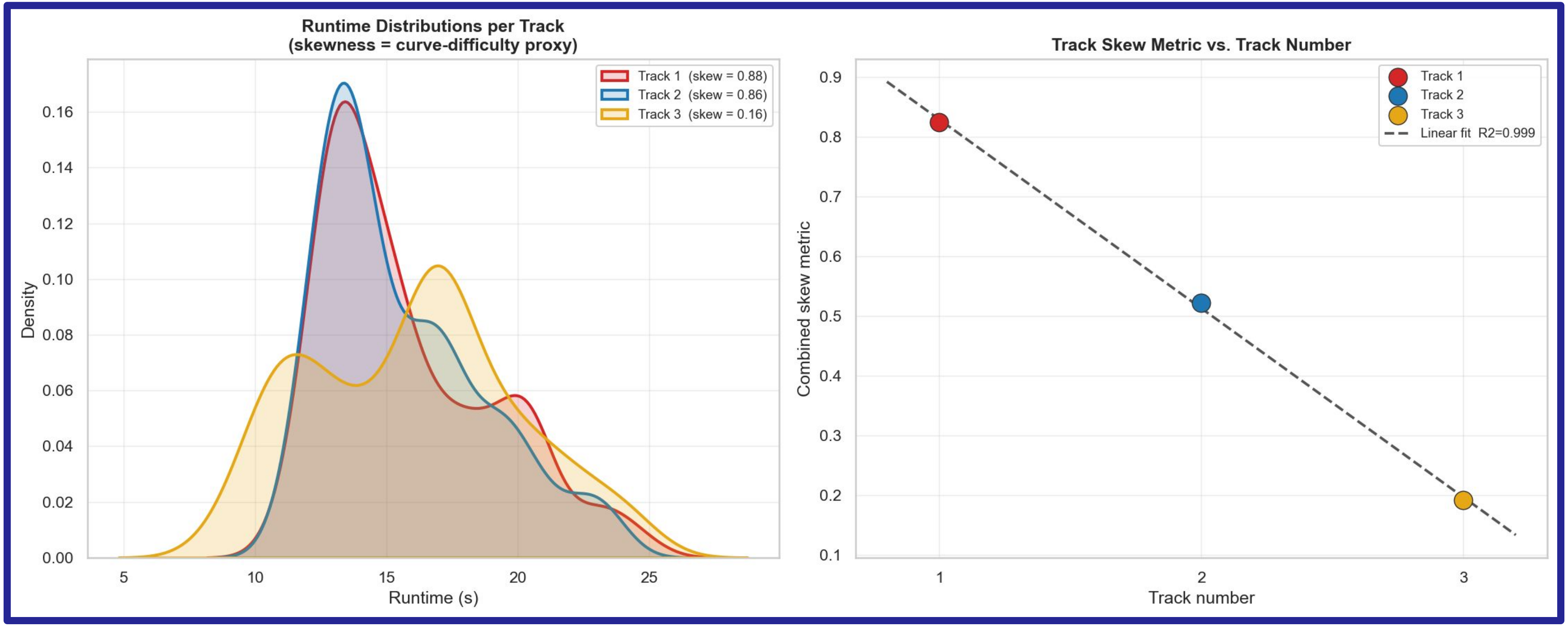
Track 3



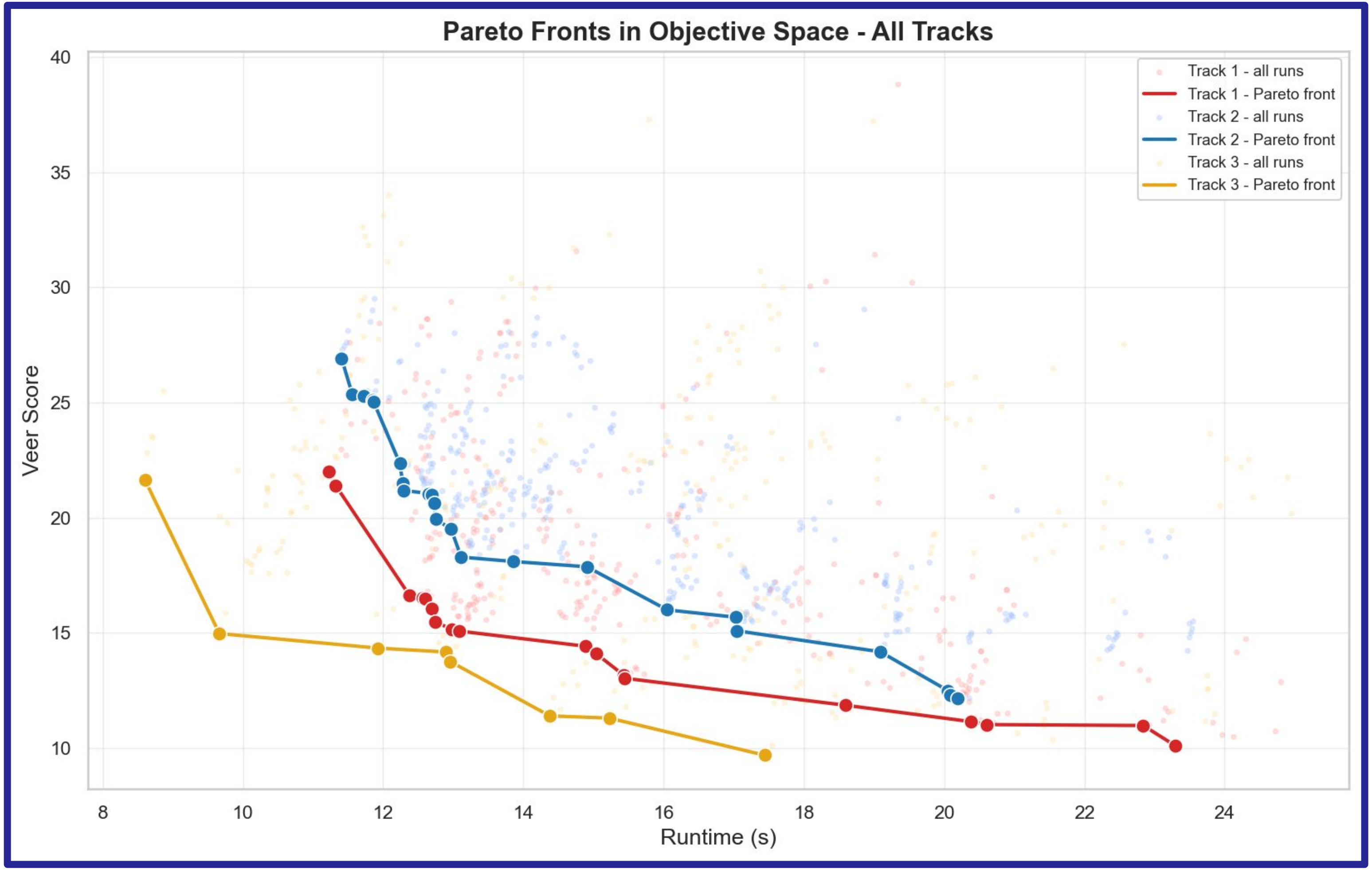
Entry curve of about 73°

Results

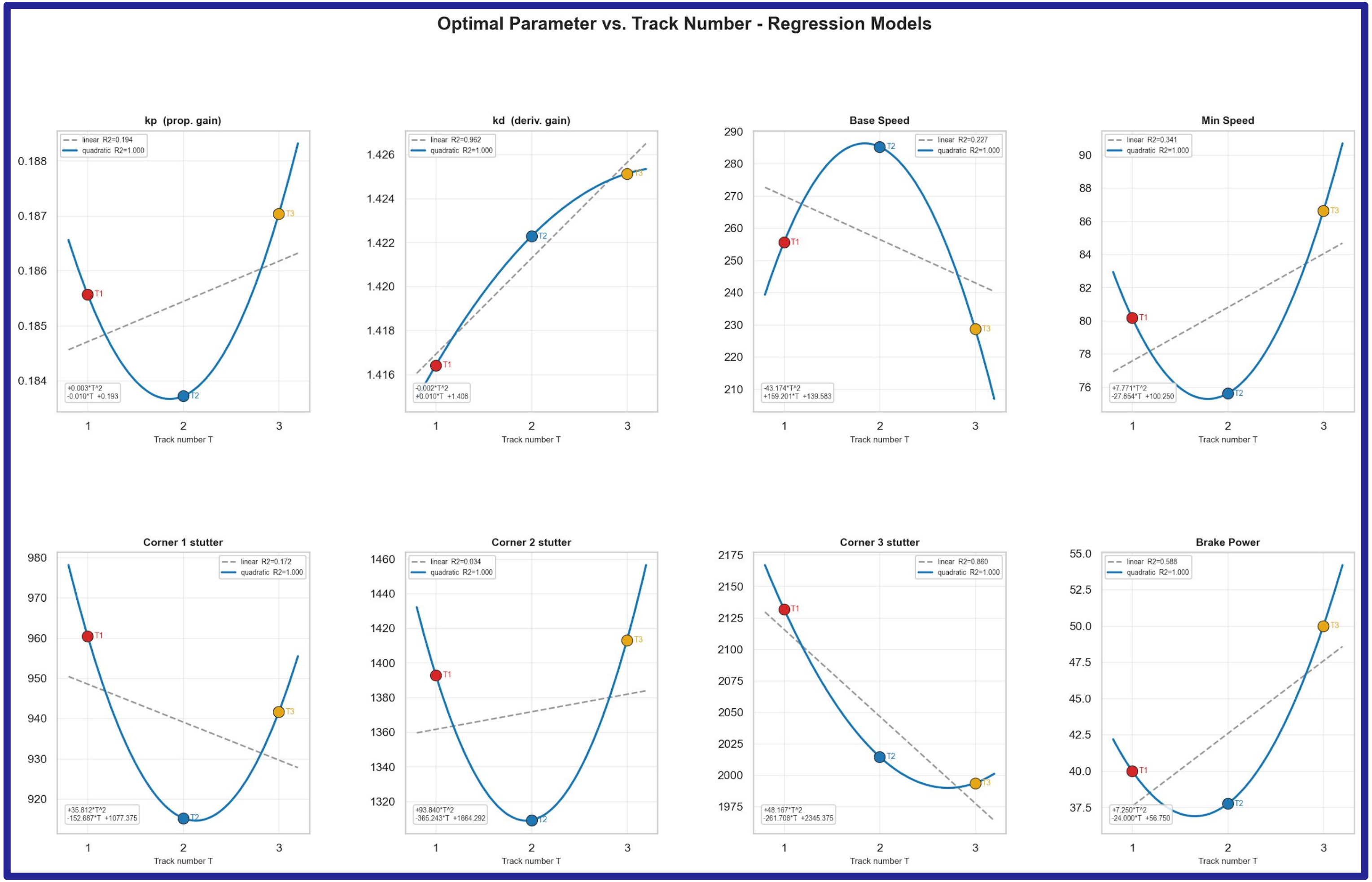
- Tracks with higher entry angles were more difficult to optimize
- Despite this NSGA was able to optimize these higher entry angle tracks successfully



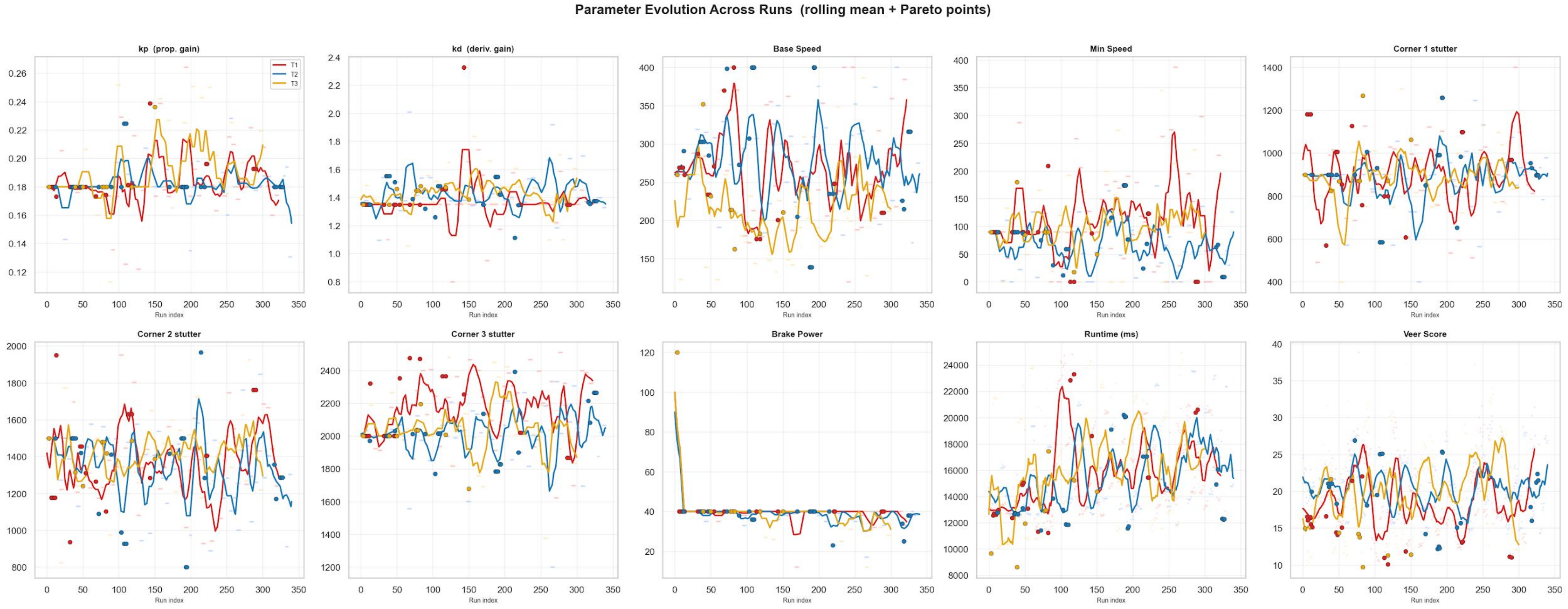
Results Cont.



Results Cont.



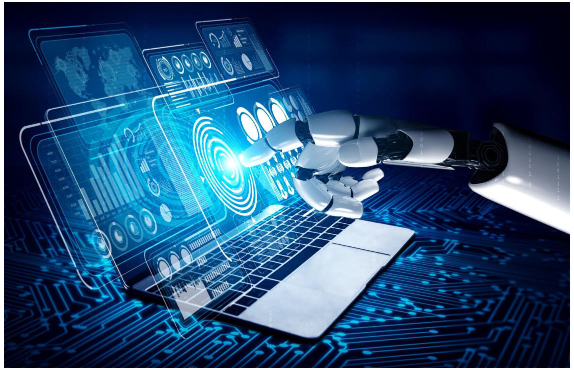
Results Cont.

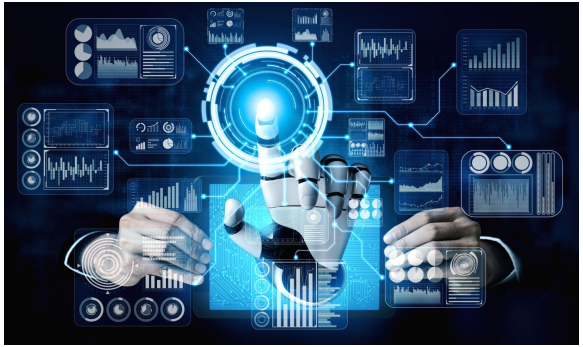


Future Work

Future improvements:

- More track variations
- More generations and trials
- Add obstacles and intersections
- Compare with reinforcement learning
- Use NSGA-III framework (more diversity)





References

Ahrari, A., Elsayed, S., Sarker, R., Essam, D., & Coello Coello, C. A. (2021). Weighted pointwise prediction method for dynamic multiobjective optimization. *Information Sciences, 546*, 349–367. https://doi.org/10.1016/j.ins.2020.08.015

Chen, R., Wu, B., Wang, H., & Huang, F. (2025). A reinforcement learning-based two-stage NSGA-II for dynamic flexible job shop scheduling problem with transportation constraints considering participators' perception. *Expert Systems with Applications, 302*, 130418. https://doi.org/10.1016/j.eswa.2025.130418

Geng, S., Ming, F., Li, D., Guo, W., Wang, L., & Wu, Q. (2025). Multiform-based evolutionary framework for large-scale constrained multi-objective optimization. *Egyptian Informatics Journal, 32*, 100854. https://doi.org/10.1016/j.eij.2025.100854

Kuo, R. J., Edbert, E., Zulvia, F. E., & Lu, S.-H. (2023). Applying NSGA-II to vehicle routing problem with drones considering makespan and carbon emission. *Expert Systems with Applications, 221*, 119777. https://doi.org/10.1016/j.eswa.2023.119777

Liu, E., Lin, Z., Chen, H., Jia, D., Liu, Y., Guo, J., Li, T., & Wei, T. (2025). Multiobjective eco-driving speed optimisation with real-time traffic: Balancing fuel, NOx, and travel time. *Energy, 324*, 135793–135793. https://doi.org/10.1016/j.energy.2025.135793

Zhang, D., Jiao, X., & Zhang, T. (2024). Lane-changing and overtaking trajectory planning for autonomous vehicles with multi-performance optimization considering static and dynamic obstacles. *Robotics and Autonomous Systems, 182,* 104797. https://doi.org/10.1016/j.robot.2024.104797